# Tutorial 8 Optimality condition

## MAT3007 Optimization

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# Outline

1. Optimality Conditions for Unconstrained Problems
2. Optimality Conditions for Linear Constrained Problems

# Review

1. a nonlinear optimization problem can be written as:

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ s.t. \quad & \boldsymbol{x} \in F \end{aligned}$$

   We call $F$ the feasible region and $\boldsymbol{x} \in F$ feasible solutions.

2. $\bar{\boldsymbol{x}}$ is global minimizer if for all $\boldsymbol{x} \in F, f(\bar{\boldsymbol{x}}) \leq f(\boldsymbol{x})$.

3. $\bar{\boldsymbol{x}}$ is local minimizer if there exists a neighborhood $N(\bar{\boldsymbol{x}})$ of $\bar{\boldsymbol{x}}$ s.t. for all $\boldsymbol{x} \in N(\bar{\boldsymbol{x}}) \cap F, f(\bar{\boldsymbol{x}}) \leq f(\boldsymbol{x})$.

# Optimality Conditions for Unconstrained Problems

**Theorem 1 (First-Order Necessary Condition).**

*If $f$ is continuously differentiable and $\mathbf{x}^*$ is a local minimizer of $f(\cdot)$ for an unconstrained problem, then we must have $\nabla f(\mathbf{x}^*) = 0$.*

**Theorem 2 (Second-Order Necessary Condition).**

*If $f$ is second-order continuously differentiable $\mathbf{x}^*$ is a local minimizer of $f(\cdot)$ for an unconstrained problem, then we must have*

1. $\nabla f(\mathbf{x}^*) = 0$;
2. $\nabla^2 f(\mathbf{x}^*)$ *is positive semi-definite.*

# Optimality Conditions for Unconstrained Problems

**Theorem 3 (Second-Order Sufficient Condition).**

*If $f$ is second-order continuously differentiable. If $\mathbf{x}^*$ satisfies:*

1. $\nabla f(\mathbf{x}^*) = 0$;
2. $\nabla^2 f(\mathbf{x}^*)$ *is positive definite.*

*Then $\mathbf{x}^*$ is a local minimizer of $f$.*

# Optimality Conditions for Unconstrained Problems

**Tips:**

1. Check if a matrix is positive definite/semi-definite;
   $det(A) = \prod_i \lambda_i$; $tr(A) = \sum_i \lambda_i$
2. When to check sufficiency/necessity
   - Find candidates for optimal solutions or prove a point x is not a local optimum
   - Prove a point x is a local optimum
3. Understand the insights behind those conditions. (Ex. $f(x) = x^3$)

## Exercise 1

Consider the function,

$$f(x, y, z) = 8x + 12y + 4xz + 2xy + x^2 - 2y^2 + z^2.$$

- Find all the local minimizers/maximizers and the global minimizer/maximizer.

## Solution for Exercise 1

Compute the gradient

$$\nabla f(x, y, z) = [8 + 4z + 2x + 2y,\ 12 - 4y + 2x,\ 4x + 2z]^T.$$

Solve $\nabla f(x, y, z) = 0$, we get $x = 14/5, y = 22/5, z = -28/5$. However, the hessian matrix is indefinite

$$\nabla^2 f = \begin{bmatrix} 2 & 2 & 4 \\ 2 & -4 & 0 \\ 4 & 0 & 2 \end{bmatrix},$$

$\sum \lambda_i = tr(A) = 0, \prod \lambda_i = det(A) = 40 > 0$. $\nabla^2 f$ has both positive and negative eigen values. This reveals that $(14/5,\ 22/5,\ -28/5)$ is neither a minimum or maximum, but a saddle point.

## Exercise 2

Find all local minimizer, local maximizer and saddle points of $f$.

$$f(x) = x_1^4 + 2(x_1 - x_2)x_1^2 + 4x_2^2$$

## Solution for exercise 2

$$\nabla f(x) = \begin{pmatrix} 4x_1^3 + 6x_1^2 - 4x_1x_2 \\ -2x_1^2 + 8x_2 \end{pmatrix}, \nabla^2 f(x) = \begin{pmatrix} 12x_1^2 + 12x_1 - 4x_2 & -4x_1 \\ -4x_1 & 8 \end{pmatrix}$$

- **Step 1** Calculate all stationary points of f by solving $\nabla f(x) = 0$ :
  $\nabla f(x) = 0 \Rightarrow x_1^* = (0,0), x_2^* = (-2,1)$.
- **Step 2** Determine the definiteness of the Hessian $\nabla^2 f(x^*)$ to decide whether the stationary points $x^*$ are local minima, maxima or saddle points.

$$\nabla^2 f(x_1^*) = \begin{pmatrix} 0 & 0 \\ 0 & 8 \end{pmatrix} \nabla^2 f(x_2^*) = \begin{pmatrix} 20 & 8 \\ 8 & 8 \end{pmatrix}$$

$tr(\nabla^2 f(x_2^*)) = 28 > 0$, $det(\nabla^2 f(x_2^*)) = 160 - 64 > 0$, $\nabla^2 f(x_2^*)$ is PD and $x_2^*$ is a local minimum.

$\nabla f(x_1^*)$ has eigen value $0, 8$, is a PSD matrix. So we can not use the second-order optimality conditions to decide whether it is a saddle point or local minimum.

We consider the function f directly around $x^*$:

$$f(|t|, 0) = t^4 + 2(|t|)t^2 = |t|^3(|t| + 2) > f(0, 0) = 0$$

$$f(-|t|, 0) = t^4 + 2(-|t|)t^2 = |t|^3(|t| - 2) < f(0, 0) = 0$$

So $x_1^*$ is a saddle point.

# Outline

1 Optimality Conditions for Unconstrained Problems

2 Optimality Conditions for Linear Constrained Problems

# Optimality Conditions for Linear Constrained Problems

The linear constrained optimization problem is

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & A\mathbf{x} \geq (=)\ b. \end{aligned}$$

**Theorem 4 (Equality Constraint).**

*If $\mathbf{x}^*$ is the minimizer, then there must exist $\mathbf{y}$ such that $A^\top \mathbf{y} = \nabla f(\mathbf{x}^*)$.*

**Theorem 5 (Inequality Constraint).**

*If $\mathbf{x}^*$ is the minimizer, then there must exist $\mathbf{y} \geq 0$ such that*

$$\begin{aligned} & A^\top \mathbf{y} = \nabla f(\mathbf{x}^*), \\ & y_i(a_i^\top \mathbf{x}^* - b_i) = 0, \forall i. \end{aligned}$$

# Optimality Conditions for Linear Constrained Problems

**Tips:**

1. Check the sign in the constraint ($=$ or $\geq$);
2. Carefully check the matrix transpose;
3. Understand the derivation of the two theorems:
   - Find the feasible direction set and descent direction set;
   - Write the condition $S_F \bigcap S_D = \emptyset$;
   - Use dual feasibility to construct the alternative systems;

## Exercise 3

Give the following problem:

$$\begin{aligned} \min_{x_1, x_2} \quad & (x_1 - 3)^2 + (x_2 - 2)^2 \\ \text{s.t.} \quad & 2x_1 + x_2 - 6 \leq 0 \\ & x_1 + 2x_2 - 6 \leq 0. \end{aligned} \tag{1}$$

- Prove that $x_1 = 2.2$ and $x_2 = 1.6$ is a local minimizer.

## Solution for Exercise 3

From the necessary condition, if $x_1^*$, $x_2^*$ is a local minimum of the problem, then there exists some $\mathbf{y} \geq 0 = [y_1; y_2]$ satisfying

$$[2x_1 - 6; 2x_2 - 4] = \begin{bmatrix} -2 & -1 \\ -1 & -2 \end{bmatrix} [y_1; y_2],$$

$$y_1(2x_1 + x_2 - 6) = 0,$$

$$y_2(x_1 + 2x_2 - 6) = 0.$$

Given that $x_1 = 2.2$ and $x_2 = 1.6$, we have the following linear system on $\mathbf{y}$:

$$[-1.6;\ -0.8] = \begin{bmatrix} -2 & -1 \\ -1 & -2 \end{bmatrix} [y_1;\ y_2],$$

$$y_2 = 0.$$

Thus $y_1 = 0.8$ and $y_2 = 0$. Hence, it satisfies the necessary condition, but we still cannot prove it is a local minimum. In fact, in this question, $f$ is convex, so it is the local minimizer (and indeed a global minimizer)

## Acknowledgement

Thank you for your time!